Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/08/2021
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2017	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Nahra, Henry K	**Contact Phone:**	216-433-5385
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hasan, Mohammad Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX17AK98G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR'S NOTE: Continuation of "Flow Boiling and Condensation Experiment (FBCE)," grants NNX13AB01G and NNX12AK14G, which covered from 6/1/2012-5/31/2017. Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the crewed mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer		

Task Description:	mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	NASA's future space missions are expected to increase greatly in both scope and complexity, and require enormous increases in propulsion power. It is presently widely acknowledged that traditional chemical propulsion systems are performance plateaued. Therefore, new energetic propulsion technologies promising very high power will be needed to achieve faster travel and greatly expanded deep space reach, while reducing specific mass (kg/kWe). Several propulsion technologies are presently under consideration, including Solar Electric Propulsion (SEP) and fission-based Nuclear Thermal Propulsion (NTP) and Nuclear Electric Propulsion (NEP). Nuclear fission options are expected to alleviate key shortcomings of SEP by greatly increasing power and decreasing specific mass. Present goals include development of a multi-megawatt fission power system with nearly one half the specific mass attainable with present liquid metal power cycles. Achieving this goal will require increasing reactor operating temperature and efficiency, as well as increasing heat rejection temperature to minimize radiator area. These goals will require detailed understanding of the influence of reduced gravity on liquid metal two-phase fluid flow and heat transfer, supported by predictive design tools that are simply unavailable today. Another R&D (research & development) need that is crucial to future NASA space missions is reducing specific mass of all space vehicle subsystems. This includes the Thermal Control System (TCS) responsible for controlling the temperature and humidity of the operating environment inside vehicles (and planetary bases). Unlike current single-phase liquid TCS technologies, which rely entirely on sensible heat rise of the working fluid to remove the heat, two-phase systems capitalize upon both sensible and latent heat to achieve orders of magnitude enhancement in heat transfer performance and therefore greatly decrease specific mass compared to single-phase counterparts. Understanding the influence of reduced gravity on two-phase fluid physics and heat transfer is therefore crucial to the development of a two-phase TCS. Unfortunately, much of the two-phase fluid flow and heat transfer know-how amassed over nearly a century of research comes from experiments that were conducted in Earth gravity. These research needs are the basis for this joint project between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and NASA Glenn Research Center (GRC). The Flow Boiling and Condensation Experiment (FBCE)is expected to provide microgravity data and video records that are vital to the development of predictive tools for space propulsion and thermal control systems. FBCE will include two test modules, one for flow boiling experiments and the other for condensation measurements. This grant has culminated in final construction of FBCE in preparation for insertion into the Fluid Integrated Rack (FIR) onboard the ISS. The PU-BTPFL team has also developed predictive mechanistic and computational models for both flow boiling and flow condensation in variable gravities.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Mudawar I, Hasan MM, Nahra HK, Balasubramaniam R, Hall NR, Guzik M. "Flow Boiling and Condensation Experiment (FBCE): from Initial Concept to Full Implementation on the International Space Station." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Abstracts for Journals and Proceedings	Lee J, Devahdhanush VS, Mudawar I, Hasan MM, Nahra HK, Balasubramaniam R, Gonia PT, May RL, Mackey JR. "Experimental, Computational, Theoretical and Analytical Investigation of Flow Boiling in Reduced Gravity." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Articles in Peer-reviewed Journals	Qiu Y, Garg D, Kim S-K, Mudawar I, Kharangate CR. "Machine learning algorithms to predict flow boiling pressure drop in mini/micro-channels based on universal consolidated data." International Journal of Heat and Mass Transfer. 2021 Oct;178:121607. https://doi.org/10.1016/j.ijheatmasstransfer.2021.121607 , Oct-2021
Awards	Mudawar I. "ASME (American Society of Mechanical Engineers) Allan Kraus Thermal Management Medal, October 2021." Oct-2021
Awards	Mudawar I. "Physical Sciences Panel Member, Congressionally requested National Academies Decadal Survey on Life and Physical Sciences for 2023-2032 NASA-sponsored space research, 2021." Oct-2021